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Fire-retardant sp^3 boron-based single ion conducting polymer electrolyte for safe, high efficiency and dendrite-free Li-metal batteries

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ABSTRACT

Single lithium ion conducting polymer electrolytes (SIPEs) are an emerging class of alternative polymer electrolytes for protecting lithium metal anode. This work explores a fully aromatic sp^3 boron based SIPE, lithium poly (4,4'-dihydroxydiphenyl sulfone borate), Li-PSB, via a one-step synthetic procedure. A series of highly porous SIPE membranes, defined as *po*-PBSBs, comprising of Li-PSB and a fully aromatic polybenzimidazole (PBI) binder are firstly prepared. As a polymer electrolyte, *po*-PBSB exhibits high lithium ion transference number of close to unity, high ionic conductivity, excellent thermal dimensional stability and promising flame-retardant. X-ray Photoelectron Spectroscopy (XPS) and Density Functional Theory (DFT) calculations depict multi-coordinated lithium ion transport channels in the *po*-PBSB membranes. By serving as polymer electrolyte in lithium metal batteries (LMBs), Li/LiFePO₄ (LFP) cell, the *po*-PBSB shows an effective suppression of lithium dendrite growth. As a consequence, *po*-PBSB based Li/LFP cell demonstrates superior cycling performance remaining 76.1% of its initial capacity with nearly 100% coulombic efficiency at 1.0C after 200 cycles. The excellent performance may be ascribed to the remarkable lithium ion transference number, high organic solvent absorption in pores and thermal and electrochemical stabilities of the *po*-PBSB. We believe that the novel SIPE materials have great potential for application in high-safety LMBs.

1. Introduction

As the high expectation of energy density in consumer electronics, electric vehicles and smart grids are being enhanced, the development of the energy storage systems with advanced energy density is getting more and more urgent [1–3]. Recently, lithium metal batteries (LMBs) have been widely considered as the most potential candidates for advanced energy storage systems due to the high theoretical specific capacity (3860 mA h g⁻¹), low density (0.59 g cm⁻³), and the lowest redox potential (−3.04 V vs. SHE) [4,5]. However, they are still suffering from several huge challenges for the practical commercial application, including lithium dendrites, the thermodynamic instability of lithium metal (unstable solid electrolyte interface, SEI), and the volume and morphological changes during cycling [6–8]. In particular, the uncontrollable lithium dendrite growth during repeated lithium plating/stripping process will pierce the separator, resulting in serious safety issues, such as internal battery short circuits, thermal runaway, and even

explosion [9–12].

Lithium ion transference number, t_{Li}^+ , is one of the key parameters of the electrolyte, directly regulating to the concentration polarization during charge/discharge process, causing formation of concentration gradients, and depleting the supply of Li⁺ at the lithium electrode/electrolyte interface upon lithium plating [13–15]. According to the Equation: $\Delta c \approx \frac{I(1-t_{Li}^+)L}{FD}$, where, Δc is the steady-state concentration gradient in the electrolyte, t_{Li}^+ stands for the Lithium ion transference number, L represents for electrolyte thickness, F and D are Faraday's constant and active charge carrier diffusion coefficient in the electrolyte, respectively, a higher t_{Li}^+ can significantly suppress formation of concentration polarization of the electrolyte [16]. Unfortunately, the dual-ion small lithium salt, such as LiBOB [17,18], LiBF₄ [19], LiTFSI [20], and LiPF₆ [21], commonly possess a t_{Li}^+ below 0.3, giving rise to serious uneven lithium deposition on the surface of lithium metal anodes, and finally leading to lithium dendrite growth.

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